

Willingness to Interact: Social Resources Facilitate Pulling Actions toward Social Avatars in Virtual Reality

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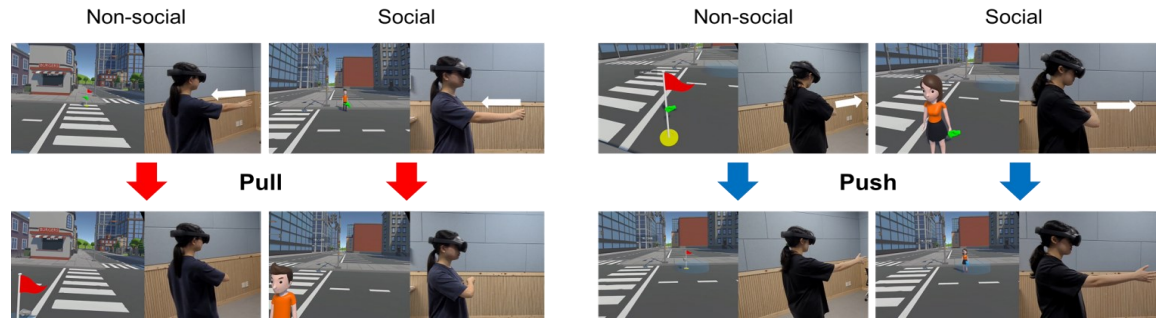


Figure 1: Study settings. The images illustrate the movement of non-social (flag) and social (male and female avatars) targets as participants perform pulling (left two columns) and pushing (right two columns) actions using bare-hand gesture interactions.

ABSTRACT

This study examined whether individuals with richer real-world social resources exhibit a greater willingness to interact with social avatars in Virtual Environment (VE). Employing a psychological approach, we assessed willingness to interact by measuring response times for pulling or pushing avatars. The results revealed that participants with richer social resources performed faster pulling actions toward social avatars, indicating a heightened willingness to engage. Notably, these effects were specific to social targets (i.e., avatars) and were not observed with non-social targets (i.e., a flag).

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality; Applied computing—Law, social and behavioral sciences—Psychology

1 INTRODUCTION

Individuals possess diverse traits; some actively seek social interactions, while others prefer to limit them. To encourage the use of metaverse systems—emerging as promising alternatives to real-world social spaces—these systems can be customized to individual traits by adjusting the level and frequency of required social interactions. This raises a key question: which user characteristics should be prioritized to effectively customize these Virtual Environments (VE)?

Among various human characteristics, this study focuses on people's social resources, a critical indicator of their willingness to engage with new encounters in real-world settings [1]. Previous research has consistently shown that individuals' social resources profoundly influence their behaviors and decisions regarding social

interactions [2, 7]. Given the robust association between social resources and the willingness to interact in the real world, this paper answers whether this link persists in a VE with an avatar. Instead of assessing all possible types of interactions in VE, we adopted a psychological method to assess the willingness to interact with a target (e.g., an avatar). Specifically, we focus on measuring pulling and pushing behaviors, actions that reflect unconscious attitudes toward a target [3]. This phenomenon, known as the Approach-Avoidance Tendency (AAT) [9], reveals that pulling is indicative of approach behaviors while pushing suggests avoidance [3, 8]. The response times associated with these actions serve as indicators of an individual's psychological willingness to interact, offering a nuanced understanding of social interaction dynamics in VE [4, 10].

In this paper, we examine the relationship between real-world social resources and the response times for pulling and pushing an avatar, which indicates the level of the willingness to interact with avatars. To the best of our knowledge, this is the first study to validate the relationship between real-world social resources and the willingness to engage with avatars. Investigating this relationship in a VE could offer valuable insights for designing customized VE in the metaverse, such as determining the optimal number of interacting agents for users of metaverse system.

2 METHODOLOGY

We conducted a user study to examine the relationship between social resources and response times for pulling and pushing an avatar in VE. Given that shorter response times indicate greater interest and willingness to engage [8, 10], we hypothesized that individuals with richer real-world social resources will exhibit faster pulling actions toward social avatars in VE. The study received approval from Chonnam National University Institutional Review Board (1040198-231122-HR-174-02).

2.1 Experimental Design

Participants were recruited through email, poster advertisement, and a website. A total of 103 participants (51 women and 52 men, ages ranging from 19 to 31, $M = 22.89$, $SD = 2.60$) were enrolled. The system utilized the Unity engine (2022.3.13f1) with the Meta XR All-in-One SDK (version 59.0.0). Participants used the Meta Quest Pro Head-Mounted Display (HMD) for hand-tracking. The

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VE featured an urban landscape with targets participants to pull or push. The study employed a between-subject design where participants were assigned to perform either pulling or pushing actions. In the pulling condition, targets moved from public to personal space, and in the pushing condition, from personal to public space, following Hall's interpersonal space framework [6]. All actions were performed bare-handed, without the use of handheld devices.

To examine the influence of gender on social motivation, male and female avatars were prepared for both same- and opposite-sex interactions. To mitigate strong emotional responses that could affect response times, we selected non-photorealistic avatars as social stimuli. Avatars were set at a height of 1.3 m, shorter than all participants, to minimize intimidation. The order of avatar presentation was counterbalanced using a Latin Square design. Additionally, a non-social stimulus, a flag, was included to provide a clear contrast with the social avatars.

2.2 Procedure and Data Collection

After providing consent, participants completed a pre-questionnaire, adapted from earlier research [5], to assess their social resources. This questionnaire included three indicators: frequency of participation in social gatherings or meetings, number of friends, and membership in groups or organizations.

Participants stood in a designated area, watched an instructional video, and practiced pulling or pushing actions at least 10 times with a virtual tree to familiarize themselves with the process. The main task involved three targets: a non-social flag and two social avatars. Each participant performed 25 repetitions of pulling or pushing actions for each target (a total of 75 times). Response times, defined as the cumulative time for 25 repetitions per target, were recorded for analysis.

3 RESULTS AND DISCUSSION

A simple moderation analysis was conducted using the PROCESS macro (10,000 bootstrapped samples). Social resources served as the independent variable, response time as the dependent variable, and action type as the moderator. The effect of social resources on response times varied with action type for both same-sex and opposite-sex avatars (same-sex: b (SE) = 3.88 (1.69), p = .024; opposite-sex: b (SE) = 4.22 (1.81), p = .022). Significant main effects of social resources on response time were also found (same-sex: b (SE) = -7.35 (2.75), p = .009; opposite-sex: b (SE) = -7.21 (2.95), p = .016). As shown in Fig. 2, participants with richer social resources exhibited significantly faster response times when pulling avatars (same-sex: b (SE) = -3.48 (1.25), p = .007; opposite-sex: b (SE) = -2.99 (1.34), p = .029), but not when pushing (same-sex: p = .726; opposite-sex: p = .313). For the non-social target, however, the relationship between social resources and response times did not vary with action type (p = .152), and no significant effect of social resources on response time was observed (p = .187).

Consistent with our hypothesis, these participants indeed pulled both same- and opposite-sex avatars more rapidly. The effect was particularly pronounced for pulling (approach) actions, but not for pushing (avoidance) actions, presumably because approach behaviors are socially overt and explicit, making them easier to capture. In contrast, avoidance behaviors are often more implicit, potentially influenced by social desirability, and may not be as effectively measured through pushing actions in VE. These results indicate that willingness to interact with social avatars in VE is closely linked to their social resources, suggesting that individuals with richer social resources are more open and positive toward new social encounters in the metaverse. Additionally, our findings reveal that faster pulling interactions with richer social resources are specific to social stimuli (same- and opposite-sex avatars) and do not extend to non-social stimuli (a flag).

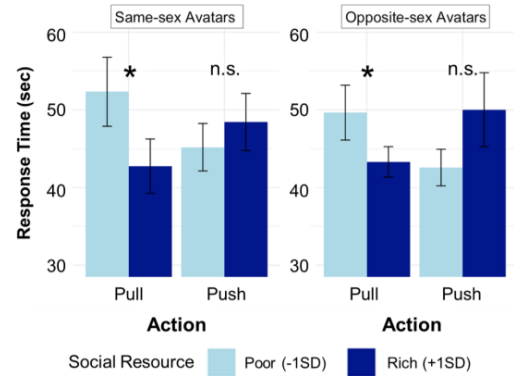


Figure 2: Interaction effect of Action (Pulling vs. Pushing) and Social Resources (Poor vs. Rich) on Response Time. The asterisks indicate significant interaction (* = p < .05).

4 CONCLUSION

This study illustrates how real-world social resources influence user attitudes toward social avatars in VE. Results indicate that individuals with richer social resources are more inclined to engage with social avatars, offering key insights for designing future metaverse systems that promote natural and meaningful interactions.

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